

NAMAN MANOCHA

AI ENGINEER — Applied ML, RAG & Production AI Systems

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SUMMARY

AI/ML engineer who ships end-to-end research products spanning data pipelines, retrieval and model logic, backend APIs, and deployment. Independently developed FundersAI, a research platform combining deterministic financial metrics, hybrid retrieval, and explainable ML clustering, and CareFlow Intelligence, an agentic RAG and fine-tuning system. Currently developing AI products independently. Core technologies: Python, FastAPI, Next.js, Supabase/PostgreSQL, Docker, and Kubernetes.

TECHNICAL SKILLS

ML & AI: LangGraph, RAG, Hybrid Search, pgvector, ChromaDB, K-Means Clustering, Logistic Regression, MLflow, scikit-learn, Prompt Engineering, Hugging Face, Langfuse

Backend & Cloud: Python, FastAPI, REST APIs, SQL, PostgreSQL, Supabase, Redis, Docker, Kubernetes, GitHub Actions, Google Cloud Run

Web & Data: Next.js, React, TypeScript, ETL Pipelines, Web Scraping, Data Validation

EXPERIENCE

Artificial Intelligence Intern — Internmo (Remote)

Sept 2026 – Present

- **Vantara:** built churn and 90-day customer-value pipeline on 1M+ retail transactions with leakage-safe features, SHAP/LIME, and FastAPI/Streamlit/Docker. FlowCast: building traffic, congestion, and accident-risk forecasts (PyTorch LSTM/GRU). Northgate: building stock return forecasts with FinBERT sentiment and a max-Sharpe optimizer. Client-style project briefs.

Technology Manager — PayGain Multiservices Pvt. Ltd.

Feb 2025 – Sept 2026

Kolkata, India

Scoped and designed a company-wide mobile app (design stage) — client/employee onboarding, document access, product activation, and payments.

Designed and deployed a lead-capture automation (Make) triggered on Google Form submissions; also drove 30+ client conversions by building and managing the company website and digital discovery flows.

INDEPENDENT PROJECTS

FundersAI — AI-Powered Research Platform for Indian Stocks & Mutual Funds

fundersai.co.in

Python, FastAPI, Next.js, Supabase/pgvector, LangGraph, Groq, OpenAI, Cloudflare R2 | ~144K source LOC, 856 files, 442 commits, 178 test files — OpenAI Build Week | Live: www.fundersai.co.in | Repo: github.com/Naman6019/FundersAI

- **Citation-Grounded RAG:** built a seven-stage LangGraph workflow with retrieval grading, query rewriting, citation-aware synthesis, and explicit abstention when evidence is insufficient.
- **Explainable ML:** fund-similarity engine using cosine similarity and k-means clustering over 13 standardized features, plus a logistic-regression triage model (MLflow) for parser-review prioritization.
- **Data Engineering:** automated collection and normalization of official mutual-fund disclosures, with data-quality checks and targeted normalizer fixes for affected records.

CareFlow Intelligence — Agentic RAG & Fine-Tuning System

Python, FastAPI, Next.js, Supabase/pgvector, ChromaDB, Unsloth, Ollama | ~7,300 LOC, non-clinical, synthetic-data-only

- **Agentic RAG + Ingestion:** built a citation-grounded document agent with automated ingestion from OpenFDA, PubMed, clinical-guideline, and media sources.
- **RAFT Fine-Tuning:** built a training-data pipeline with reference evidence, hard distractors, and abstention cases; fine-tuned Qwen2.5-1.5B via 4-bit QLoRA (rank 16, alpha 16), quantized it to GGUF, and served it locally via Ollama with human approval gates for data intake.

OTHER PROJECTS

TalentOS (All Things Agentic Hackathon, Taskmaster Track): dual-pipeline opportunity platform using Google ADK, LangGraph, and Firestore to collect ~2,900 postings/run, apply deterministic pre-filtering that eliminates 89.5% before any LLM call, and generate human-reviewable application materials, backed by a 373-test hermetic offline suite (~11s). FundersAI Reports: decoupled LangGraph report-synthesis service hosted on K3s. SQuAD QA Benchmarking: fine-tuned and benchmarked BERT, BiDAF, and DistilBERT; published as a research paper. GitResume: self-built evidence-backed resume CLI (PyPI: git-resume-agent, ~4K LOC, 28 commits) that validates managed claims before compiling Word/PDF output.

EDUCATION & ACHIEVEMENTS

B.Tech, Computer Science Engineering (AI & ML Specialization) — VIT Amaravati

June 2021 – Sept 2025

CGPA: 8.2/10 | Published research paper on QA/NLP | Certifications: Azure Administrator Associate, AWS Certified Cloud Practitioner